AVIKAM JHA

RTL Design & Micro-architecture Engineer
Tempe, AZ | (480) 200-3712 | avikamjh@asu.edu | linkedin.com/in/avikamjha

SUMMARY

Computer Engineering M.S. candidate (**GPA 4.0**, Dec 2026) focused on **RTL design and micro-architecture**. Designed a **5-stage pipelined 32-bit CPU** with full forwarding, five stall conditions, atomics, and a CAM, and architected a **sparse GCN inference accelerator** taken through synthesis and place-and-route to **zero setup and hold violations across 2,196 paths** at **5.26 mW**. Optimizes RTL for power, performance, area, and timing, co-writes verification plans, and brings **2+ years as a software engineer** with Python and Shell automation.

EDUCATION

Arizona State University, Tempe, AZ | M.S., Computer Engineering | GPA 4.0/4.0 | Dec 2026

Relevant coursework: VLSI Design, Digital Systems and Circuits, SoC Design, Computer Architecture II, Digital Verification and Test

National Institute of Technology, Delhi | B.E., Electronics and Communication Engineering | July 2022

TECHNICAL SKILLS

HDL & Design Tools: SystemVerilog, Verilog, ModelSim, Xilinx Vivado (simulation), Synopsys Design Compiler, Cadence Innovus, Cadence Virtuoso, Synopsys HSPICE

Architecture & Modeling Tools: gem5, CUDA

Scripting & Languages: Python, Tcl, Cadence SKILL, Shell/Bash, Make, C, C++

Platforms & PDKs: Linux, Git, Kubernetes; ASAP7 7nm FinFET, SAED 32nm

PROJECTS

5-Stage Pipelined 32-bit CPU | SystemVerilog | Aug – Dec 2025

- Built a 32-bit RISC-style core from memory, register file, ALU, and PC primitives to a full pipeline with **ID/EX, EX/MEM, and MEM/WB** registers and a hazard-detection control FSM.
- Designed a **forwarding unit** with EX/MEM-over-MEM/WB priority and three independent bypass networks (ALU operands, branch comparands, jump-register targets), resolving **five stall conditions** across load-use, jump-register, and branch hazards.
- Extended the ISA with **load-linked/store-conditional** atomics and built a **CAM** for single-cycle cache tag lookup.

Sparse GCN Inference Accelerator: Architecture, RTL & Timing Closure | SystemVerilog, Design Compiler, Innovus, ASAP7 7nm | 2025

- Architected a **three-stage transformation, aggregation, and argmax** datapath, replacing dense adjacency multiplication with **COO sparse-format** accumulation that only fetches the feature rows an edge touches.
- Cut switching in a **96-element dot-product unit** with zero-operand early termination and added **per-block clock gating**, reaching **5.26 mW** and **88.5 ns** latency against a 100 ns budget at 16,590 cells in 0.054 mm^2.
- Took the RTL through synthesis and place-and-route to **zero setup and hold violations across 2,196 paths**, and re-ran one testbench at RTL, post-synthesis, and post-layout netlists to separate timing mismatches from functional bugs.

VeriChip: Verification Plans & Functional Coverage | SystemVerilog, C | 2025

- Co-authored specification-mapped test plans, verified a 4-state FSM and all **16 ALU commands**, and closed coverage with **four covergroups** and 3-way crosses against a C-based PLI reference model.

Cache Replacement Policy & Out-of-Order Study | gem5, C++, Python | 2025

- Implemented the **LRU-IPV** PseudoLRU policy from published research as a C++ SimObject in gem5, keeping strict recency ordering with a shared per-set priority vector at O(associativity) update cost.
- Swept fetch/decode/issue/commit widths and measured a **1.7x** slowdown from 8-wide to 2-wide out-of-order pipelines, identifying memory access as the bottleneck.

Registered Adder, Full Flow Bring-Up | Design Compiler, Innovus, Virtuoso, ASAP7 7nm | 2025

- Ran a boundary-flopped datapath through synthesis, power stripes, CTS, APR, and GDS import into Virtuoso for DRC and LVS, with inputs on the west edge and outputs on the east edge.
- Adapted the **Tcl synthesis and APR scripts** for a multi-module hierarchy, reported core and chip area, and analyzed the top three worst-case setup and best-case hold paths from post-route timing reports.

Full-Custom 4-bit Adder, PPA Optimization | Virtuoso, HSPICE, IC Validator, SAED 32nm | 2024

- Designed a mirror-architecture adder and reached **240 ps**, **27.43 uW**, and **23 um²** by upsizing only the final six stacked transistors 1.5x instead of the whole circuit, earning a top score on the course power-delay-area metric.

RTL IP & SoC Program | In Progress, Sept – Dec 2026

- Writing lint-clean, parameterized IP with design specs (UART, synchronous and **asynchronous FIFOs**, round-robin arbiter, **CDC synchronizers**, Hamming ECC) and integrating a **RISC-V RV32I SoC** over AXI4-Lite and APB (target: Dec 2026).

EXPERIENCE

Park+ (Parviom Technologies Pvt. Ltd.), Gurugram, India | SDE 2 | Mar 2022 – Jul 2024

- Owned **Linux and Kubernetes infrastructure** for a **60+ microservice** platform across networking, security, and compliance, and built **Shell and Python automation** that cut issue-response time by **20%**.

Arizona State University | Graduate Service Assistant | Aug 2025 – May 2026

- Evaluated EEE 120 digital design labs (adders, multiplexers, decoders, ALU, counters, FSMs) and graded EEE 202 circuit analysis, checking CAD and SPICE simulations for correctness and timing.

Arizona State University | Research Aide, Introduction to Materials Science | May – Aug 2025

- Authored **40+ assignment questions** and technical documentation across 10+ semiconductor topics for a 120+ student cohort.